

MA1521 L1 Week 05 Mon 20260907

Find the differential (dw or dz). Make the answer look as neat as possible.

a) $w = \ln(xyz)$ b) $w = x^3y^2z$ c) $z = \frac{x-y}{x+y}$

The following equations define w implicitly as a function of the other variables. Find dw in terms of all the variables by taking the differential of both sides and solving algebraically for dw .

$$\text{a) } \frac{1}{w} = \frac{1}{t} + \frac{1}{u} + \frac{1}{v} \qquad \text{b) } u^2 + 2v^2 + 3w^2 = 10$$

In the following, find $\frac{df}{dt}$ for the composite function $f(x(t), y(t), z(t))$ in two distinct ways:

(i) use the chain rule, then express your answer in terms of t by using $x = x(t)$, etc.;

(ii) express the composite function f in terms of t , and differentiate.

a) $w = xyz, \quad x = t, \quad y = t^2, \quad z = t^3$

b) $w = x^2 - y^2, \quad x = \cos t, \quad y = \sin t$

c) $w = \ln(u^2 + v^2), \quad u = 2 \cos t, \quad y = 2 \sin t$

- a) Use the chain rule for $f(u, v)$, where $u = u(t)$, $v = v(t)$, to prove the product rule

$$D(uv) = v Du + u Dv, \quad \text{where } D = \frac{d}{dt}.$$

- b) Using the chain rule for $f(u, v, w)$, derive a similar product rule for $\frac{d}{dt}(uvw)$, and use it to differentiate $te^{2t} \sin t$.

Let $w = f(x, y)$, and make the change of variables $x = u^2 - v^2$, $y = 2uv$. Show

$$(w_x)^2 + (w_y)^2 = \frac{(w_u)^2 + (w_v)^2}{4(u^2 + v^2)}$$

a) Let $w = f(y/x)$; i.e., w is the composite of the functions $w = f(u)$, $u = y/x$.

Show that w satisfies the PDE (partial differential equation)

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = 0$$

b) Let $w = f(x^2 - y^2)$; show that w satisfies the PDE $y \frac{\partial w}{\partial x} + x \frac{\partial w}{\partial y} = 0$.

c) Let $w = f(ax + by)$; show that w satisfies the PDE $b \frac{\partial w}{\partial x} - a \frac{\partial w}{\partial y} = 0$.